

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a7d75acf392e608d3.jsonl`  
- SHA-256: `102a676cf8178f2fa96b0fd4b02dd33fe0abc7c577560ee56a813ebd249a1157`  
- Source modified: `2026-06-06T17:37:50+00:00`  
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- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:36:32.129Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"A two-stage audio pipeline (energy-based peak detection followed by a small CNN) can detect racket-ball contact events from sound with near-millisecond temporal precision (energy peaks detected on average 0.2 ms after onset, even with noise), demonstrating that audio is a high-resolution local cue for hit timing in a racket sport."

****Source:**** <https://arxiv.org/html/2409.11760v1> (primary)
****Supporting quote:**** "The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring... we use energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:36:34.633Z)

I'll verify this claim about the audio pipeline for racket-ball contact detection. Let me start by examining the primary source.

3. assistant (2026-06-06T17:36:38.298Z)

Let me fetch the primary source to examine the actual claims in the paper.

4. user (2026-06-06T17:36:54.713Z)

Analysis of Table Tennis Bounce Sound Detection Paper

****Sport:**** Table tennis (ping pong)

****Audio Detection Pipeline:**** The system detects when a ball bounces on a racket and classifies the bounce to identify racket type and spin applied.

****Temporal Precision Claims:**** The paper states that "the detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring" in noisy conditions, with 0.09 ms precision in clean environments.

****What is Detected:**** The pipeline detects ball-racket contact events (bounces), not general hits. The authors note that table tennis ball-racket contact duration ranges from 1.3-1.8 milliseconds, making precision critical.

****Two-Stage Pipeline:****

1. ****Energy peak detection**** uses an exponential moving average threshold with a high-pass filter (10 kHz cutoff) to identify potential bounces with minimal latency
2. ****CNN classification**** refines detections and categorizes them by surface type and spin

presence

****Reported Accuracy:**** The CNN classifier achieved:

- Spin classification: 0.98 precision, 0.95 recall, 0.96 F1-score
- Surface classification: 0.97 precision, 0.97 recall, 0.97 F1-score

****Racket Sport:**** Yes, table tennis is a racket sport.

5. assistant (2026-06-06T17:36:58.239Z)

The source supports the core technical claims. Let me verify some details more precisely by reading the actual paper, and then check whether there are any qualifications or contradictions I should know about.

6. user (2026-06-06T17:37:07.328Z)

Analysis of Temporal Resolution Claims

****(1) The 0.2 ms temporal resolution claim and conditions:****

"The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring. The increased delay when there is background noise is due to the peak being proportionally less pronounced."

The 0.2 ms figure was measured with human speech overlaid to simulate noise. Clean conditions achieved 0.09 ms.

****(2) How the 0.2 ms was measured:****

The paper states this was tested on "test data" but provides no explicit description of ground-truth measurement methodology. The authors compare frame energy to a moving exponential average threshold empirically, suggesting computational latency rather than validation against independent timing reference.

****(3) Limitations and qualifications:****

"The audio signal is first filtered using a 5th-order Butterworth high-pass filter with a cutoff frequency of 10 kHz." The method depends critically on this filtering step. The authors note their approach "works effectively in a noiseless environment" but acknowledge "other sounds can degrade the detector's performance."

****(4) Recording setup:****

"The different bounce sounds were recorded with the Zoom H4n Pro Handy Recorder, a directional microphone, at 44.1kHz" with "distances ranging from 50 cm to 2 m." This represents controlled lab recordings, not match footage.

****(5) Detection scope:****

The peak detection captures racket contact bounces. The dataset includes table bounces as separate classification targets, suggesting detection applies broadly to energy peaks meeting threshold criteria.

7. assistant (2026-06-06T17:37:10.768Z)

I now have a clear picture. Let me assess the claim against the checklist. The key question for the research context is whether this generalizes ? let me check if there's any contradicting evidence about audio hit detection in real match conditions, since the claim's framing ("audio is a high-resolution local cue for hit timing in a racket sport") implies broader applicability.

8. user (2026-06-06T17:37:18.903Z)

Web search results for query: "audio racket-ball contact detection match footage limitations

false positives badminton table tennis sound"

Links: [{"title": "(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis", "url": "https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"}, {"title": "A comprehensive review of ball detection techniques in sports", "url": "https://peerj.com/articles/cs-3079.pdf"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/26119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://arxiv.org/abs/2409.11760"}, {"title": "EventAnchor: Reducing Human Interactions in Event Annotation of Racket Sports Videos", "url": "https://arxiv.org/pdf/2101.04954"}, {"title": "Wearable Audio and IMU Based Shot Detection in Racquet Sports", "url": "https://arxiv.org/pdf/1805.05456"}, {"title": "Context modulates the impact of auditory information on visual anticipation", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9346043/"}, {"title": "Racket Sports Sound Effects - Tennis, Badminton & Squash | SONNISS", "url": "https://sonniss.com/sound-effects/racket-sports-sound-effects-library/"}, {"title": "TOTNet: Occlusion-Aware Temporal Tracking for Robust Ball Detection in Sports Videos", "url": "https://arxiv.org/pdf/2508.09650"}]

I've found several research articles on audio-based racket-ball contact detection in racquet sports. Here are the key findings:

Audio Detection Systems for Racquet Sports

Supervised classification methods are employed to detect important audio symbols including ball hits, and refinement procedures have been developed to reduce false rally scenes through further audio analysis in table tennis, badminton, and tennis contexts.

Limitations and False Positives

Table tennis ball detection models address shortcomings with existing systems like high false-positive rates. Additionally, false positives, hazy balls, and court lines remain challenges in badminton stroke detection systems.

Sound Characteristics and Detection

In table tennis, brief, high-frequency sounds produced during racket-ball impacts carry information about the racket type, the surface contacted, and whether spin was applied. Real-time-capable pipelines combine energy-based peak detection with convolutional neural networks trained on datasets of bounce samples recorded across different racket configurations.

Performance Metrics

Table tennis shot detection using energy peak detection-based approaches achieves an F-score of 81%, while padel-related algorithms show satisfying results with an average F1-score of 92%.

Underutilized Audio Potential

Audio from videos is insufficiently utilized, though audio information could be used for event recognition by detecting the sound of ball contact through stroke detection.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:37:37.019Z)

I have enough to make my assessment. Let me apply the adversarial checklist.

****Checklist analysis:****

1. ****Is the claim supported by the quote?**** Yes, closely. The paper (arXiv:2409.11760v1) is

about table tennis (a racket sport), describes exactly a two-stage pipeline (energy-based peak detection ? CNN classification), and states "detecting energy peaks on average no later than 0.2 ms after occurring." Clean conditions: 0.09 ms; with speech-noise overlay: 0.2 ms. The CNN F1 scores (0.96-0.97) are real. The claim's wording is faithful.

2. ****Contradicting evidence?**** No source disputes that audio gives high temporal resolution for racket-ball contact ? multiple papers corroborate audio as a strong hit-timing cue (tennis, padel F1?92%, table tennis). The corroborating literature actually strengthens, not refutes, the technical claim.

3. ****Source quality?**** Primary arXiv source matches the claim. The claim is not extraordinary ? millisecond-level peak detection in filtered audio is plausible and corroborated.

4. ****Outdated?**** No ? Sept 2024 paper.

5. ****Marketing/cherry-picked?**** It's an academic paper, not marketing.

****However ? the legitimate skeptical caveats (qualifications, not refutations):****

- The 0.2 ms figure was measured in a ****controlled lab setup**** (Zoom H4n Pro directional mic, 50cm-2m, single ball/racket, synthetic speech noise) ? NOT in-the-wild match footage with crowd noise, reverb, or multiple simultaneous sounds. The paper itself admits "other sounds can degrade the detector's performance."

- The "0.2 ms" is detection ****latency relative to the peak****, and depends critically on a 10 kHz high-pass filter that exploits the brief high-frequency impact signature ? a property that holds for table tennis but is not directly transferable to badminton (shuttle "thwack" has different spectral content), the asker's actual target sport.

These caveats qualify the claim's **generalizability** to the user's badminton-match context, but they do NOT refute the claim ****as stated****, which is carefully scoped to "a racket sport" and is accurately backed by the cited primary source.

The claim is accurate, current, well-sourced, and faithfully represents the primary paper. It does not overstate beyond what the paper shows (it says "a racket sport," not "all racket sports" or "match footage"). I refute it = false, with documented caveats.

10. user (2026-06-06T17:37:45.484Z)

Structured output provided successfully

11. assistant (2026-06-06T17:37:50.882Z)

I've completed the verification and submitted my structured output.

Verdict: refuted = false (high confidence). The claim is well-supported by the cited primary source (arXiv:2409.11760v1), accurately represents the two-stage pipeline and the 0.2 ms temporal-resolution figure, is current (Sept 2024), and is corroborated rather than contradicted by independent literature.

The important caveats I flagged (which qualify but do not refute the claim as worded): the 0.2 ms figure comes from a controlled lab setup with synthetic noise, not real match footage, and it relies on table-tennis's high-frequency impact signature via a 10 kHz high-pass filter ? properties that don't directly transfer to badminton shuttle audio in real match conditions, which is the user's actual target.